

CloudFlow Analytics: screening and capital decisions under uncertainty

Your guide for the lab. Start with the email, then work each card in order. Everything else is optional reference.

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Submit

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The request

Subject: **Two decisions under uncertainty: I need numbers I can defend**



Chief Financial Officer

To: Senior Business Risk Analyst (you)

Cc: Executive Risk Committee

Reply by Tue, Sep 29 · 11:59 PM ET

Hi Risk Analytics team,

We have two decisions in front of the Risk Committee: whether to adopt an automatic screening rule for customer churn, and how to invest in capacity. Our current plans lean on averages and gut feel. I want each answer tied to evidence, with units and denominators, and a plain statement of what the evidence can't tell us.

Specifically, I need answers to 5 questions:

1. Should we screen high-ticket accounts?

Part A

Customer Success wants to flag accounts with heavy support-ticket volume for high-touch support. Using last year's 1,000 accounts, what is our baseline churn rate, and how much higher is it among high-ticket accounts? Someone told me "80%." Please check the denominator. And tell me plainly whether this shows screening would *prevent* churn.

2. Large Hub or Modular Hub?

Part B

Demand next year could be high, medium or low, and each hub pays off differently in each case. Which one has the higher expected value, by how much, and what would make us choose the other?

3. Is our profit plan too optimistic?

Part C

The plan assumes average demand. Simulate 1,000 demand scenarios. Does profit at average demand overstate what we should expect? How bad is a bad year (the 5th percentile)?

4. How will you measure downside?

Checkpoint

The Committee cares about how often we lose money, not only the average. Show me the measure you'll use.

5. Option A or Option B for the next three years?

Take-home

A costs \$1.0M up front; B costs \$0.9M and is steadier. Using an 8% discount rate, compare expected NPV and the chance of losing money, recommend one, and tell me what would reverse your recommendation.

Please send me:

- your completed workbook, and
- a one-page memo that leads with the recommendation, cites every number with its unit and denominator, and discloses any AI use.

The account data and simulation inputs are attached. Thanks. I'm counting on this for the Committee.

Best,

Chief Financial Officer



cloudflow_churn_cohort.csv



reference_trials.csv



LAB02_Learning_Workbook.xlsx

Your lab at a glance

Before class

setup
untimed

Part A

8 min
Slide 24 → 25

Part B

10 min
Slide 42 → 43

Part C

12 min
Slide 65 → 66

Checkpoint

3 min
Slide 76 → 80

Take-home

45 min
At home

Submit

checklist
Due Tue, Sep
29

Before class: get ready (untimed)

1

Unzip **LAB02_Student_Package.zip** into one folder.

Keep every file together. SPSS looks for the CSV files next to the syntax.

2

Open **LAB02_Learning_Workbook.xlsx** and go to the **Start Here** sheet.

Type your first name, last name and NetID (the part before @nyu.edu). Use **File > Save As** with the workbook name it shows.

3

Open **LAB02_SPSS.sps** in SPSS and fix the first line.

Replace **/REPLACE/WITH/LAB02_FOLDER** with the path of your unzipped lab folder (the folder, not the **.sps** file; see [Find your folder path](#)). Save As the **.sps** name shown on Start Here.

4

Select the whole **H0** block, then choose **Run > Selection**.

The Output Viewer opens with **H0 Import and Data Screen**.



Ready when: the H0 output shows the expected case count and your workbook is saved under your own name.



Stuck? Don't start Part A on the timer. Ask for help first, or see the SPSS environment guide.

Find your folder path (for the `CD` line)

SPSS needs the path of the **folder** you unzipped, the one that holds `LAB02_SPSS.sps` and the CSV files. It is **not** the path to the `.sps` file. If you open the folder and see the `.sps` file and the CSV files inside, that is the right folder.

Windows

1. Unzip first: right-click `LAB02_Student_Package.zip` > **Extract All** > **Extract**. Do not work inside the zip.
2. Open the extracted `LAB02_Student_Package` folder, then the **Student** folder inside it. You should now see `LAB02_SPSS.sps` and the CSV files. This is your lab folder.
3. Click once in the address bar at the top of File Explorer: it turns into the path. Press **Ctrl+C**.
4. Or: go up one level, hold **Shift**, right-click that folder > **Copy as path**, then delete the double quotes it adds.

Looks like

```
C:\Users\you\Downloads\LAB02_Student_Package\Student
```

Mac

1. Unzip first: double-click `LAB02_Student_Package.zip` in Finder.
2. Open the extracted `LAB02_Student_Package` folder, then the **Student** folder inside it. You should now see `LAB02_SPSS.sps` and the CSV files. This is your lab folder.
3. Go up one level and click that folder **once** to select it (the folder, not a file inside it). Press **Option+Command+C** (Copy as Pathname), or right-click it, hold **Option** and choose **Copy ... as Pathname**.

Looks like

```
/Users/you/Downloads/LAB02_Student_Package/Student
```

Paste it between the single quotes on the `CD` line:

```
CD 'C:\Users\you\Downloads\LAB02_Student_Package\Student'.  
CD '/Users/you/Downloads/LAB02_Student_Package/Student'.
```

* *Windows*

* *Mac*

⚠ **Common mistakes:** the path ends in `/LAB02_SPSS.sps` (remove the file name); the quotes or the final period are missing; the files are still inside the zip; OneDrive or iCloud has not finished downloading the folder. If H0 says a file cannot be found, check the path first. If the path itself contains an apostrophe (`'`), put the path in double quotes instead, for example `CD "/Users/you/Kim's files/LAB02_Student_Package/Student".`

How every card works

🟡 Yellow cells: you type here 🟣 Lavender cells: guidance

✓ **Check your work** box on each sheet turns to "✓ Checked"

Each card opens with a short recap of the session content it uses, with the slides and Toolkit pages to reread. Type each formula starting with `=`, then press **Enter**. The cells are already formatted, so never multiply a percentage by 100.

Part A · Do high-ticket accounts churn more?

From slide 24 → work this card → Back to slide 25

· bring: The exact conditional churn risk percentage of 35.3 percent for high-ticket accounts.

WHAT THE SESSION COVERED · SECTION 1 PROBABILITY RULES AND CONDITIONAL BUSINESS RISK

A **contingency table** sorts every account by two categories at once. The inside cells are *joint* counts; the row and column totals are *marginal* counts. From one table you can read four kinds of probability:

- **Marginal:** one event across everyone. $P(A) = \frac{\text{row or column total}}{N}$
- **Joint:** both events together, across everyone. $P(A \cap B) = \frac{\text{cell count}}{N}$
- **Union:** either event (or both). Subtract the overlap so it isn't counted twice. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- **Conditional:** one event *within* a group. The group's total becomes the denominator. $P(A | B) = \frac{P(A \cap B)}{P(B)}$

Bayesian updating is the same idea, used to revise a baseline risk (the *prior*) after you see a signal. **Base-rate neglect** is the classic error: because most accounts renew, even a signal that catches 80% of churners flags more renewers than churners. So $P(H | C)$ and $P(C | H)$ are very different numbers.

COURSE SLIDES

Slides 10–24

Anatomy of Two-Way Contingency Tables 11 · General Addition Rule of Probability 12–14 · Calculating Marginal and Joint Probabilities 15 · Bayes Theorem and the Law of Total Probability 17 · Bayes Theorem for Binary Business States 18–19 · Updating Churn Risk for CloudFlow Analytics 20

CONCISE TOOLKIT

pp. 4–10

Bayesian Updating p. 6 · CloudFlow Analytics p. 7

FULL TOOLKIT

pp. 4–21

Bayesian Updating p. 11 · CloudFlow Analytics p. 12

THE EQUATIONS YOU ARE BUILDING

C = account churned · H = account had high support tickets · $n(\cdot)$ = number of accounts

$$P(C | H) = \frac{P(C \cap H)}{P(H)} = \frac{n(C \cap H)}{n(H)} = \frac{B9}{D9}$$

SAME ANSWER BY BAYES' RULE

$$P(C | H) = \frac{P(H | C)P(C)}{P(H | C)P(C) + P(H | \bar{C})P(\bar{C})}$$

FIRST, THE SETUP · ONCE PER SPSS SESSION

Run this **once each time you open SPSS**, before any part. On the **CD** line, replace **/REPLACE/WITH/LAB02_FOLDER** with the path of your **unzipped lab folder** (keep the single quotes and the final period). It is the folder that holds **LAB02_SPSS.sps** and the data files, **not** the **.sps** file itself: see **Find your folder path**. Select from the first line down to the end of the H0 block and choose **Run > Selection**. Already ran it since you opened SPSS? Skip to the block for this part.

- * *Lecture connection: Descriptive Frequency Tables .*
- * *Why: Import the exact cohort as strings and inspect row counts and allowed category values .*
- * *How: GET DATA reads the exact CSV basename .*
- * *How: String formats preserve AccountID and category labels; blank strings are declared missing .*
- * *Read the result: Inspect H0 for 1,000 cases and only recorded Yes or No category values .*
- * *Watch out: Do not let SPSS infer AccountID as numeric .*
- * *Run: Open the supplied syntax through File > Open > Syntax .*
- * *Run: Select the complete H0 block and choose Run > Selection .*
- * *Run: Switch to Output Viewer and locate H0 Import and Data Screen .*

CD '/REPLACE/WITH/LAB02_FOLDER'. * ← change only this path

* *Name the output block so you can find it in the Viewer.*

TITLE 'H0 Import and Data Screen'.

* *Read the data file into SPSS (file type, name, and each variable's format).*

GET DATA

/TYPE=TEXT

/FILE='cloudflow_churn_cohort.csv'

/ENCODING='UTF8'

/ARRANGEMENT=DELIMITED

/FIRSTCASE=2

/DELCASE=LINE

/DELIMITERS=', '

/QUALIFIER='\"'

/VARIABLES=

AccountID A32

Churn A3

HighTickets A3.

* *Keep a working copy in memory so later passes are fast.*

CACHE.

* *Run the pending transformations now.*

EXECUTE.

* *Give the open dataset a name so later commands can point to it.*

DATASET NAME CohortData.

* *Declare which codes mean "missing" so they are left out of statistics.*

MISSING VALUES Churn HighTickets ('').

* *Attach readable descriptions to variables (they appear in output tables).*

VARIABLE LABELS

AccountID 'Recorded CloudFlow account identifier'

Churn 'Recorded churn outcome'

HighTickets 'Recorded high-ticket indicator'.

* *Frequency table: how many cases fall in each category.*

FREQUENCIES VARIABLES=Churn HighTickets

/ORDER=ANALYSIS.

BEFORE THE TIMER · INSTRUCTOR DEMO

Follow along: in SPSS, run the **H1** block and leave **H1 Conditional Churn Risk** open.

SPSS BLOCK FOR THIS PART · ALREADY IN YOUR SYNTAX FILE

You do not type this: it is already in `LAB02_SPSS.sps`, the one syntax file you use all session. Once per SPSS session, run the `CD` line and the H0 block first (the setup code is above). Then press **Ctrl+F**, search `TITLE 'H1`, select from the first comment line down to the last period, and choose **Run > Selection**. Green lines start with `*`: they are comments that explain the command below them, and SPSS ignores them. Every command ends with a period.

* *Lecture connection: Bayesian Updating .*

* *Why: Cross-tabulate Churn by HighTickets using observed category pairs .*

* How: Rows represent Churn and columns represent HighTickets .

* How: Column percentages use observed Churn within each ticket column .

* Read the result: Read the Churn=Yes count and column percentage within HighTickets=Yes .

* Watch out: Do not report the row percentage as posterior churn risk .

* Run: Select the complete H1 block after H0 has run .

* Run: Choose Run > Selection .

* Run: Locate H1 Conditional Churn Risk in the Viewer .

TITLE 'H1 Conditional Churn Risk'.

* Two-way table: counts and percentages for one variable by another.

```
CROSSTABS
  /TABLES=Churn BY HighTickets
  /FORMAT=AVALUE TABLES
  /CELLS=COUNT ROW COLUMN TOTAL
  /COUNT ROUND CELL.
```

OPEN

Open IN-CLASS_Lab Part A!A7 .

FILL IN THE YELLOW CELLS

#	Cell	Type this	In math	Check
1	B7	=ROWS(Cohort[AccountID])	N_{rows} all source rows	1,000
2	B9	=COUNTIFS(Cohort[HighTickets],\$A9,Cohort[Churn],B\$8) Fill to C10.	$n(H \cap C)$ and the other 3 cells	96, 176, 24, 704
3	D9	=SUM(B9:C9) Fill to D10.	$n(H)$ row totals	272
4	B11	=SUM(B9:B10) Fill to D11.	$n(C)$, N column totals	120, 880, 1,000
5	B13	=B11/D11	$P(C)$ baseline churn	12.0%
6	B14	=D9/D11	$P(H)$ high-ticket share	27.2%
7	B15	=B9/D11	$P(C \cap H)$ both	9.6%
8	B16	=B13+B14-B15	$P(C \cup H)$ either	29.6%
9	B17	=B9/B11	$P(H C)$ among churners	80.0%
10	B18	=B9/D9	$P(C H)$ among high-ticket	35.3%
B20: pick the best statement from the dropdown.				
11	B20	B21: 2-3 sentences: state the numerator, denominator, comparison with baseline, and association-only limitation.		
	B21			
	B25	B25: caption the SPSS H1 Conditional Churn Risk output. 1-2 sentences: name H1 Conditional Churn Risk, identify the count and column percentage read, and state the Excel agreement check.		

✅ **Done when:** B18 displays the conditional churn risk as **35.3%**, using high-ticket complete pairs as the denominator. Save the same personalized workbook and resume at IN-CLASS_Lab Part B!A7 . **Save.**

⚠️ **Most common mistake:** reporting **80%** as the churn risk. That is $P(H \mid C)$ in B17, the share of *churners* who had high tickets. The CFO asked for $P(C \mid H)$: of the 272 high-ticket accounts, how many churned?

$$P(C \mid H) = \frac{96}{272} = 0.353 \neq P(H \mid C) = \frac{96}{120} = 0.800$$

Stuck? Three quick fixes

- **#NAME?** The data table must be named `Cohort` with headers `AccountID` , `Churn` , `HighTickets` . Don't rename them.
- **Shows 3,529%**. You multiplied by 100. Remove the `*100` ; the cell is already a percentage.
- **Four counts don't add up to D11**. Blank cells are missing, not "No". Keep the criteria exactly `Yes` and `No` .

Why does 80% shrink to 35.3%? (Bayes' rule worked through)

$$P(C \mid H) = \frac{\frac{P(H \mid C)}{0.80 \times 0.12} \times \frac{P(C)}{0.20 \times 0.88}}{\frac{P(H \mid C)}{0.80 \times 0.12} + \frac{P(C)}{0.20 \times 0.88}} = \frac{0.096}{0.272} = 0.353$$

churners flagged renewers flagged

The signal flags 96 churners but also 176 renewers, because renewers are 88% of the base. High-ticket accounts churn at about three times the 12.0% baseline. That supports screening them first. It does not show that screening would *reduce* churn.

Part B · Large Hub or Modular Hub?

From slide 42 → work this card → Back to slide 43

· bring: The optimal strategy selection and its expected value premium over the alternative.

WHAT THE SESSION COVERED · SECTION 2 EXPECTED MONETARY VALUE AND DECISION ANALYSIS

A decision separates what you **control** (the alternatives: Large Hub, Modular Hub) from what you **don't** (the states of nature: high, medium, low demand). States must be mutually exclusive and cover every possibility, so their probabilities sum to 1.

- **Payoff table:** rows are alternatives, columns are states, and each cell is the *net* payoff $v(d_j, s_i)$ after capital costs.
- **Expected Monetary Value:** the probability-weighted average payoff of one alternative. In Excel this is `SUMPRODUCT(probabilities, payoffs)`.
- **Decision tree rollback:** work right to left. At each chance node (circle), compute the EMV; at the decision node (square), keep the branch with the larger EMV and prune the other.
- **EMV is not an outcome:** it's a long-run average; you will actually get one of the three payoffs. A committee that can't absorb the worst state may reject the higher-EMV branch.
- **Sensitivity:** find the probability at which the preferred alternative flips (the tipping point).

COURSE SLIDES

Slides 25–42

Mechanism of constructing a payoff table and... 27 · Expected Monetary Value Formula 28–30 · Worked evidence: CloudFlow capacity alternatives 31 · The Rollback Mechanism and Backward Induction 33 · Expected Monetary Value Calculation 34–35 · Excel Rollback Calculation with SUMPRODUCT 36–37

CONCISE TOOLKIT

pp. 11–18

Expected Monetary Value p. 12

FULL TOOLKIT

pp. 22–41

Expected Monetary Value p. 23 · CloudFlow Analytics p. 40

THE EQUATIONS YOU ARE BUILDING

d_j = alternative (hub) · s_i = demand state · $v(d_j, s_i)$ = net payoff, million dollars

$$\sum_{i=1}^3 P(s_i) = 1 \quad \text{EMV}(d_j) = \sum_{i=1}^3 P(s_i) v(d_j, s_i)$$

ROLLBACK AND PREMIUM

$$V_{\text{decision}} = \max_j \text{EMV}(d_j) \quad \text{Premium} = \text{EMV}(\text{chosen}) - \text{EMV}(\text{other})$$

Just opened SPSS? Run the setup first (the **CD** line and H0; code in Part A).

BEFORE THE TIMER · INSTRUCTOR DEMO

Follow along: in SPSS, run the **H2** block and leave **H2 Capacity EMV Audit** open.

SPSS BLOCK FOR THIS PART · ALREADY IN YOUR SYNTAX FILE

You do not type this: it is already in `LAB02_SPSS.sps`, the one syntax file you use all session. Once per SPSS session, run the `CD` line and the `H0` block first (setup code in Part A). Then press **Ctrl+F**, search `TITLE 'H2'`, select from the first comment line down to the last period, and choose **Run > Selection**. Green lines start with `*`: they are comments that explain the command below them, and SPSS ignores them. Every command ends with a period.

```
* Lecture connection: Expected Monetary Value .
* Why: Audit the probability weights and capacity EMVs using the same stated CloudFlow assumptions .
* How: Three retained cohort rows provide a reproducible workspace .
* How: Weighted payoffs are summed by capacity alternative .
* Read the result: Read the probability total, both EMVs and rolled-back node value in million dollars .
* Watch out: Do not mistake the largest state payoff for the largest EMV .
* Run: Select the complete H2 block .
* Run: Choose Run > Selection .
* Run: Locate H2 Capacity EMV Audit and read its listed values .
TITLE 'H2 Capacity EMV Audit'.
* Make a working copy so the original data stay untouched.
DATASET COPY CapacityAudit.
* Switch to the named dataset; the next commands work on it.
DATASET ACTIVATE CapacityAudit.
* Keep only the rows that meet the condition (others are deleted).
SELECT IF ($CASENUM <= 3).
* Declare new text variables.
STRING DemandState (A6).
* Start a block of commands that run only when the condition is true.
DO IF ($CASENUM = 1).
    COMPUTE DemandState='High'.
    COMPUTE StateProbability=.30.
    COMPUTE LargePayoff=12.
    COMPUTE ModularPayoff=9.
* Alternative condition inside a DO IF block.
ELSE IF ($CASENUM = 2).
    COMPUTE DemandState='Medium'.
    COMPUTE StateProbability=.50.
    COMPUTE LargePayoff=3.
    COMPUTE ModularPayoff=5.
ELSE.
    COMPUTE DemandState='Low'.
    COMPUTE StateProbability=.20.
    COMPUTE LargePayoff=-9.
    COMPUTE ModularPayoff=1.
* Close the DO IF block.
END IF.
* Create or overwrite a variable with a formula, row by row.
COMPUTE LargeWeighted=StateProbability*LargePayoff.
* Create or overwrite a variable with a formula, row by row.
COMPUTE ModularWeighted=StateProbability*ModularPayoff.
* Collapse rows into group summaries (one row per group).
AGGREGATE
    /OUTFILE=* MODE=ADDVARIABLES
    /ProbabilityTotal=SUM(StateProbability)
    /LargeHubEMV=SUM(LargeWeighted)
    /ModularHubEMV=SUM(ModularWeighted).
* Create or overwrite a variable with a formula, row by row.
COMPUTE RolledBackValue=MAX(LargeHubEMV,ModularHubEMV).
* Set how values display (width and decimals); the stored values do not change.
FORMATS StateProbability ProbabilityTotal (F5.3)
```

```

LargePayoff ModularPayoff LargeHubEMV ModularHubEMV RolledBackValue (F8.2).
* Make the next transformation apply to the next procedure only.
TEMPORARY.
* Keep only the rows that meet the condition (others are deleted).
SELECT IF ($CASENUM=1).
* Print the listed variables row by row.
LIST VARIABLES=ProbabilityTotal LargeHubEMV ModularHubEMV RolledBackValue.
* Close a dataset you no longer need.
DATASET CLOSE CapacityAudit.
* Switch to the named dataset; the next commands work on it.
DATASET ACTIVATE CohortData.

```

OPEN

Open **IN-CLASS_Lab Part B!A7**.

FILL IN THE YELLOW CELLS

#	Cell	Type this	In math	Check
1	B7:D7	30% 50% 20%	$P(s_H), P(s_M), P(s_L)$ must total 100%	
2	B10:D10	12 3 -9	$v(\text{Large}, s_i)$ \$ millions	
3	B11:D11	9 5 1	$v(\text{Modular}, s_i)$ same order: H, M, L	
4	D14	=SUMPRODUCT(\$B\$7:\$D\$7, B10:D10) Fill to D15.	EMV(Large), EMV(Modular)	3.30, 5.40
5	D16	=MAX(D14:D15)	$\max_j \text{EMV}(d_j)$ rollback	5.40
6	D17	=MAX(D14:D15) - MIN(D14:D15)	$\text{EMV}_{\max} - \text{EMV}_{\min}$ premium	2.10
7	B18 B19 B23	B18: pick the best statement from the dropdown. B19: 2-3 sentences: state the EMV criterion, selected branch, premium, and one risk-tolerance condition that could change the choice. B23: caption the SPSS H2 Capacity EMV Audit output. 1-2 sentences: name H2 Capacity EMV Audit, identify the values read, and state the Excel agreement check.		

✅ **Done when:** D15 displays the Modular Hub EMV as **5.40 million dollars**, and D16 agrees with the larger branch EMV. Save the same workbook and resume at **IN-CLASS_Lab Part C!A7**. **Save.**

⚠️ **Most common mistake:** picking the Large Hub because it has the biggest single payoff (\$12M). The EMV rule weighs every state by its probability, and the Large Hub's -\$9M low-demand state pulls its EMV down.
 $\text{EMV}(\text{Large}) = 0.30(12) + 0.50(3) + 0.20(-9) = 3.30 < \text{EMV}(\text{Modular}) = 0.30(9) + 0.50(5) + 0.20(1) = 5.40$

Stuck? Three quick fixes

- **EMV looks wrong.** Payoffs must follow the same High, Medium, Low order as the probabilities in B7:D7.
- **Probabilities show 3000%.** Type 30% or 0.3, not 30.
- **SUMPRODUCT shifts when you copy it.** Keep the \$ signs on \$B\$7:\$D\$7 so the probabilities stay fixed.

Why might the Committee still reject the higher-EMV hub? (optional)

EMV is a long-run average. CloudFlow builds one hub and gets one outcome. If a branch's worst state would breach a liquidity or covenant limit, a risk-averse committee can rationally reject it even when its EMV is higher. Here the Modular Hub also has the safer worst case (\$1M vs. -\$9M), so both views agree.

Part C · Is profit at average demand too optimistic?

From slide 65 → work this card → Back to slide 66

· bring: The simulated valuation gap between naive base-case profit and mean stochastic profit.

WHAT THE SESSION COVERED · SECTION 3 MONTE CARLO SIMULATION AND THE FLAW OF AVERAGES

The **flaw of averages**: plugging average demand into a *nonlinear* model does not give average profit. The capacity cap and salvage rule make profit nonlinear:

- **Upside is capped**: when demand exceeds capacity, extra customers add nothing: $\text{sales} = \min(D, Q)$.
- **Downside is not**: when demand falls short, surplus units sell only at salvage: $\text{surplus} = \max(Q - D, 0)$. So expected profit falls *below* profit at mean demand.
- **Monte Carlo simulation**: runs the model many times with sampled demand. **Inverse transform sampling** turns a probability u between 0 and 1 into a demand value: `NORM.INV(u, mean, sd)`.
- **Summaries**: mean profit, sample standard deviation, and the 5th-percentile profit, which is a *dollar amount*, not a percentage.
- **More trials don't fix a wrong model**: they make the estimate more precise, not the assumptions more correct.

COURSE SLIDES

Slides 44–65

How Non-Linear Payoffs Break Symmetry 45 · Mathematical Formulation of the Flaw of Averages 46–48 · Excel Formulas for Bounded Inventory Payoffs 49–51 · CloudFlow Profit Under Demand Uncertainty 52 · Random Variate Generation in Spreadsheets 54 · Normal Inverse Transformation for Continuous... 55–57

CONCISE TOOLKIT

pp. 19–25

The Flaw of Averages p. 24

FULL TOOLKIT

pp. 42–63

The Flaw of Averages p. 43 · CloudFlow Analytics p. 62

How the lab differs from the slides: class showed `RAND()` with an Excel Data Table. In the lab you use the fixed probabilities in `Data_Trials` (column `u_001`) instead, so everyone's numbers match and can be checked. The `RAND()` version is in the optional `LAB02_Live_Exploration.xlsx`.

THE EQUATIONS YOU ARE BUILDING

D = demand · Q = capacity (1,000) · p = price (\$100) · s = salvage (\$15) · c = cost (\$40) · u_k = fixed probability for trial k

$$D_k = \mu + \sigma \Phi^{-1}(u_k) \quad (\mu = 1000, \sigma = 250) \text{Profit}_k = p \cdot \min(D_k, Q) + s \cdot \max(Q - D_k, 0) - c \cdot Q$$

WHY THE AVERAGE MISLEADS

$$E[\text{Profit}(D)] \neq \text{Profit}(E[D])$$

Just opened SPSS? Run the setup first (the `CD` line and H0; code in Part A).

BEFORE THE TIMER · INSTRUCTOR DEMO

Follow along: in SPSS, run the **H3** block and leave **H3 Reference Profit Summary** open.

SPSS BLOCK FOR THIS PART · ALREADY IN YOUR SYNTAX FILE

You do not type this: it is already in `LAB02_SPSS.sps`, the one syntax file you use all session. Once per SPSS session, run the `CD` line and the H0 block first (setup code in Part A). Then press **Ctrl+F**, search `TITLE 'H3'`, select from the first comment line down to the last period, and choose **Run > Selection**. Green lines start with `*`: they are comments that explain the command below them, and SPSS ignores them. Every command ends with a period.

```
* Lecture connection: Monte Carlo Simulation Engine .
* Why: Reproduce the in-class demand and profit summaries from shared reference probabilities .
* How: IDF.NORMAL transforms fixed u_001 values using mean 1,000 and standard deviation 250 .
* How: A temporary sorted copy uses  $h=1+(n-1)p$  and linear interpolation for the inclusive percentile .
* Read the result: Read mean profit, sample standard deviation, inclusive fifth percentile and numeric trial count .
* Watch out: Do not substitute random draws or SPSS default percentile definitions for the fixed inputs .
* Run: Select the complete H3 block, including the reference import .
* Run: Choose Run > Selection .
* Run: Locate H3 Reference Profit Summary and inspect the listed summary .
TITLE 'H3 Reference Profit Summary'.
* Read the data file into SPSS (file type, name, and each variable's format).
GET DATA
  /TYPE=TXT
  /FILE='reference_trials.csv'
  /ENCODING='UTF8'
  /ARRANGEMENT=DELIMITED
  /FIRSTCASE=2
  /DELCASE=LINE
  /DELIMITERS=', '
  /QUALIFIER=''
  /VARIABLES=
    trial F10.0
    u_001 F20.16.
* Keep a working copy in memory so later passes are fast.
CACHE.
* Run the pending transformations now.
EXECUTE.
* Give the open dataset a name so later commands can point to it.
DATASET NAME ReferenceTrials.
* Create or overwrite a variable with a formula, row by row.
COMPUTE ValidU=0.
* Compute a value only for rows that meet the condition.
IF (NOT MISSING(u_001) AND u_001 > 0 AND u_001 < 1) ValidU=1.
* Compute a value only for rows that meet the condition.
IF (ValidU=1) SimDemand=IDF.NORMAL(u_001,1000,250).
* Compute a value only for rows that meet the condition.
IF (ValidU=1) SimProfit=MIN(SimDemand,1000)*100+MAX(1000-SimDemand,0)*15-1000*40.
* Run the pending transformations now.
EXECUTE.
* Make a working copy so the original data stay untouched.
DATASET COPY ProfitSummary.
* Switch to the named dataset; the next commands work on it.
DATASET ACTIVATE ProfitSummary.
* Create or overwrite a variable with a formula, row by row.
COMPUTE ValidProfit=NOT MISSING(SimProfit).
* Sort the rows by the listed variables.
```

```

SORT CASES BY ValidProfit(D) SimProfit(A).
* Collapse rows into group summaries (one row per group).
AGGREGATE
  /OUTFILE=* MODE=ADDVARIABLES
  /NTrials=N(SimProfit)
  /MeanProfit=MEAN(SimProfit)
  /SDProfit=SD(SimProfit).
* Create or overwrite a variable with a formula, row by row.
COMPUTE Rank5=$SYSMIS.
* Create or overwrite a variable with a formula, row by row.
COMPUTE LowerRank=$SYSMIS.
* Create or overwrite a variable with a formula, row by row.
COMPUTE UpperRank=$SYSMIS.
* Start a block of commands that run only when the condition is true.
DO IF (NTrials > 0).
  COMPUTE Rank5=1+(NTrials-1)*.05.
  COMPUTE LowerRank=TRUNC(Rank5).
  COMPUTE UpperRank=MIN(LowerRank+1,NTrials).
* Close the DO IF block.
END IF.
* Create or overwrite a variable with a formula, row by row.
COMPUTE P5LowCell=$SYSMIS.
* Create or overwrite a variable with a formula, row by row.
COMPUTE P5HighCell=$SYSMIS.
* Compute a value only for rows that meet the condition.
IF (NTrials > 0 AND $CASENUM=LowerRank) P5LowCell=SimProfit.
* Compute a value only for rows that meet the condition.
IF (NTrials > 0 AND $CASENUM=UpperRank) P5HighCell=SimProfit.
* Collapse rows into group summaries (one row per group).
AGGREGATE
  /OUTFILE=* MODE=ADDVARIABLES
  /P5Low=MAX(P5LowCell)
  /P5High=MAX(P5HighCell).
* Create or overwrite a variable with a formula, row by row.
COMPUTE P5Profit=$SYSMIS.
* Compute a value only for rows that meet the condition.
IF (NTrials > 0) P5Profit=P5Low+(Rank5-LowerRank)*(P5High-P5Low).
* Set how values display (width and decimals); the stored values do not change.
FORMATS MeanProfit SDProfit P5Profit (DOLLAR14.2) NTrials (F8.0).
* Make the next transformation apply to the next procedure only.
TEMPORARY.
* Keep only the rows that meet the condition (others are deleted).
SELECT IF ($CASENUM=1).
* Print the listed variables row by row.
LIST VARIABLES=MeanProfit SDProfit P5Profit NTrials.
* Close a dataset you no longer need.
DATASET CLOSE ProfitSummary.
* Switch to the named dataset; the next commands work on it.
DATASET ACTIVATE ReferenceTrials.

```

OPEN

Open IN-CLASS_Lab Part C!A7.

FILL IN THE YELLOW CELLS

#	Cell	Type this	In math	Check
1	B14	=MIN(B7,B9)*B10+MAX(B9-B7,0)*B11-B9*B12	Profit($E[D]$) at mean demand	\$60,000
2	A32	1 Fill to A1031.	$k = 1, \dots, 1000$ trial number	
3	B32	=NORM.INV(Data_Trials!B2,\$B\$7,\$B\$8) Fill to B1031.	$D_k = \mu + \sigma\Phi^{-1}(u_k)$	654
4	C32	=MIN(B32,\$B\$9)*\$B\$10+MAX(\$B\$9-B32,0)*\$B\$11-\$B\$9*\$B\$12 Fill to C1031.	Profit $_k$	\$30,575
5	B16	=AVERAGE(C32:C1031)	Profit mean	\$51,320
6	B17	=STDEV.S(C32:C1031)	s sample SD	\$12,701
7	B18	=PERCENTILE.INC(C32:C1031,0.05)	Profit $_{5\%}$ in dollars	\$25,884
8	B19	=B14-B16	Profit($E[D]$) – Profit	\$8,680
9	B20	=COUNT(C32:C1031)	n numeric trials	1,000
10	B22 B23 B27	B22: pick the best statement from the dropdown. B23: 2-3 sentences: compare deterministic profit with fixed-reference mean simulated profit, interpret the inclusive fifth percentile in dollars, and state one model limitation; do not use optional F9 outcomes as exact evidence. B27: caption the SPSS H3 Reference Profit Summary output. 1-2 sentences: name H3 Reference Profit Summary; report its mean, sample SD, inclusive fifth percentile, and trial count; compare the SPSS results with Excel within the stated tolerance. Report any optional Python/R comparison separately, only if you actually ran and checked it.		

✓ **Done when:** B20 reports 1,000 numeric profit trials, B16:B18 contain mean, sample standard deviation and 5th-percentile profit, and the trial area contains no formula errors. Save the same workbook. **Save.**

⚠ **Most common mistake:** reporting \$60,000 as expected profit. That is profit *at* average demand. Because sales are capped at capacity and surplus sells at salvage, the average of the 1,000 profits is lower.

Profit($E[D]$) – Profit = \$8,680
 \$60,000 \$51,320

Stuck? Three quick fixes

- **#NUM! in column B.** NORM.INV needs a probability strictly between 0 and 1. Check that B32 points at Data_Trials!B2.
- **Every row shows the same demand.** You locked the input as \$B\$2. Use plain B2 so it moves down as you fill.
- **B20 is less than 1,000.** The fill stopped early. Fill B32:C32 all the way to row 1031.

Why not use RAND()? (optional)

RAND() gives new numbers every time Excel recalculates, so no two students, and no grader, would see the same results. The fixed u_001 values give everyone the same 1,000 trials, so results can be checked exactly and matched in SPSS. To watch the numbers move with F9, use the optional LAB02_Live_Exploration.xlsx ; it isn't graded.

Checkpoint · How will you measure downside?

From slide 76 → work this card → Back to slide 80

· bring: State how the evidence supports this outcome: Evaluate operational capital investment trade-offs under stochastic demand, audit simulation models, and synthesize executive findings..

WHAT THE SESSION COVERED · SECTION 4 APPLIED RISK ANALYSIS AND SIMULATION MODELING WORKSHOP

Downside risk asks two things: how *often* outcomes fall below a benchmark (here, break-even) and how *badly*.

- **Loss probability:** the share of simulated outcomes below zero. The denominator is every *numeric* outcome.
- **5th percentile:** how bad a bad year is, in dollars.
- **Standard deviation:** measures spread in both directions, good and bad. Loss probability looks only at the bad side.
- **Decision rule from the session:** if capital preservation governs, prefer the option with the lower loss probability and a safer 5th percentile, as long as expected value stays positive. If the firm can absorb losses, prefer the higher expected value.

COURSE SLIDES

Slides 66–76

Building the Simulation Architecture in Excel 67 · Excel Dynamic Cash Flow and Simulation Formulas 68–71 · Standard Versus Flexible Center Performance 72

CONCISE TOOLKIT

pp. 26–32

FULL TOOLKIT

pp. 64–83

CloudFlow Analytics p. 81

THE EQUATIONS YOU ARE BUILDING

x_k = simulated outcome in trial k · $\text{count}(\dots)$ = number of trials that meet the condition

$$P(\text{Loss}) = \frac{\text{count}(x_k < 0)}{\text{count}(x_k \text{ numeric})} \quad \text{VaR}_{5\%} = 5\text{th percentile of } x_k$$

Just opened SPSS? Run the setup first (the **CD** line and H0; code in Part A).

BEFORE THE TIMER · INSTRUCTOR DEMO

Follow along: in SPSS, run the **H4** block and leave **H4 Applied Downside Check** open.

SPSS BLOCK FOR THIS PART · ALREADY IN YOUR SYNTAX FILE

You do not type this: it is already in **LAB02_SPSS.sps**, the one syntax file you use all session. Once per SPSS session, run the **CD** line and the H0 block first (setup code in Part A). Then press **Ctrl+F**, search **TITLE 'H4'**, select from the first comment line down to the last period, and choose **Run > Selection**. Green lines start with *****: they are comments that explain the command below them, and SPSS ignores them. Every command ends with a period.

* Lecture connection: Downside Risk and Shortfall Probability .
 * Why: Demonstrate a shortfall numerator and numeric denominator with the in-class profit trials .
 * How: LossFlag identifies profit below zero .
 * How: The sum of flags is divided by the numeric profit count .
 * Read the result: Read the shortfall count, trial denominator and stored loss fraction .

```

* Watch out: Do not include missing outcomes in the denominator .
* Run: Run H3 before selecting the complete H4 block .
* Run: Choose Run > Selection .
* Run: Locate H4 Applied Downside Check in the Viewer .
TITLE 'H4 Applied Downside Check'.
* Create or overwrite a variable with a formula, row by row.
COMPUTE LossFlag=(SimProfit < 0).
* Collapse rows into group summaries (one row per group).
AGGREGATE
  /OUTFILE=* MODE=ADDVARIABLES
  /ShortfallCount=SUM(LossFlag)
  /NumericDenominator=N(SimProfit).
* Create or overwrite a variable with a formula, row by row.
COMPUTE ShortfallProbability=ShortfallCount/NumericDenominator.
* Set how values display (width and decimals); the stored values do not change.
FORMATS ShortfallCount NumericDenominator (F8.0) ShortfallProbability (PCT6.1).
* Make the next transformation apply to the next procedure only.
TEMPORARY.
* Keep only the rows that meet the condition (others are deleted).
SELECT IF ($CASENUM=1).
* Print the listed variables row by row.
LIST VARIABLES=ShortfallCount NumericDenominator ShortfallProbability.

```

OPEN

Open **IN-CLASS_Lab Part C!B1033** . Press **Ctrl+G** (Go To), type **B1033** and press **Enter**.

FILL IN THE YELLOW CELLS

#	Cell	Type this	In math	Check
1	B1033	B1033: 2-3 sentences: use fixed-reference results to identify one expected-performance metric, one downside or spread metric, and one risk-tolerance condition that could change a decision; treat F9 variation as optional live exploration only.		
	B1037	B1037: caption the SPSS H4 Applied Downside Check output. 1-2 sentences: name H4 Applied Downside Check, identify the fixed-reference shortfall numerator and numeric-trial denominator, and state what the in-class method demonstrates without selecting a take-home option.		

✅ **Done when:** **B1033:H1035** contains one evidence-based principle and one reversal condition without claiming that the take-home option has already been selected. **Save.**

⚠️ **Most common mistake:** using the standard deviation as the downside measure. SD counts good surprises as "risk" too. Loss probability and the 5th percentile look only at the bad side.

Stuck? Three quick fixes

- **Can't find B1033.** Press **Ctrl+G** (Go To), type **B1033** , press **Enter**. It sits below the 1,000 trial rows.
- **Not sure which numbers to cite.** Use mean profit (B16) for expected performance and the 5th percentile (B18) or the H4 shortfall share for downside.
- **Tempted to pick a take-home option.** Don't yet. The checkpoint states a principle; the take-home numbers decide.

Can the option with the larger expected value still be rejected? (optional)

Yes. An option with a lower expected value can be the rational choice when it clearly lowers the chance of a loss and the Committee has a stated limit on liquidity, covenants or capital preservation. You'll make that call with your own take-home numbers.

Take-home · Option A or Option B for three years?

At home → work this card → Submit

WHAT THE SESSION COVERED · SECTION 4 APPLIED RISK ANALYSIS AND SIMULATION MODELING WORKSHOP

The take-home reuses the Part C method on a **three-year capital decision**, using the three-layer model from the session: assumptions, then a one-trial engine, then 1,000 trials.

- **NPV:** discount each year's cash flow at 8%. The initial investment happens at time zero, so it is subtracted *outside* Excel's `NPV()`.
- **One shared shock per trial:** the same z_k drives both options and all three years. That makes a fair side-by-side comparison, but it assumes a persistent business condition rather than independent years. Name that limitation in the memo.
- **Compare four numbers per option:** expected NPV, standard deviation, 5th-percentile NPV and loss probability. The higher expected NPV is not automatically the answer; it depends on the Committee's risk tolerance.
- **Reversal condition:** say what would reverse your recommendation, for example a different discount rate or risk tolerance.

COURSE SLIDES

Slides 66–76

Building the Simulation Architecture in Excel 67 · Excel Dynamic Cash Flow and Simulation Formulas 68–71 · Standard Versus Flexible Center Performance 72

CONCISE TOOLKIT

pp. 26–32

FULL TOOLKIT

pp. 64–83

CloudFlow Analytics p. 81

THE EQUATIONS YOU ARE BUILDING

j = option (A or B) · I_j = initial investment · μ_j, σ_j = mean and SD of annual cash flow · $r = 0.08$

$$z_k = \Phi^{-1}(u_k) \quad CF_{j,k} = \mu_j + \sigma_j z_k \quad NPV_{j,k} = -I_j + \sum_{t=1}^3 \frac{CF_{j,k}}{(1+r)^t}$$

COMPARE THE OPTIONS ON

$$NPV_j, \quad s_j, \quad NPV_{j,5\%}, \quad P(\text{Loss}_j) = \frac{\text{count}(NPV_{j,k} < 0)}{\text{count}(NPV_{j,k} \text{ numeric})}$$

OPEN

Open POST-CLASS_Part 1!A7.

TRY IT FIRST · NEW FUNCTIONS

These functions are new in this part. Type each example in any empty cell and check you get the same result, then use the function below.

- `=ABS(-4.2)` returns **4.2**: the size of a number without its sign.
- `=COUNTIF({-3, 5, -1, 8}, "<0")` returns **2**: counts the cells that meet one condition.

- `=NORM.S.INV(0.975)` returns **1.96**: turns a probability into a standard-normal z value.
- `=NPV(10%, 100, 100, 100)` returns **248.69**: present value of three end-of-year cash flows; subtract a time-zero cost outside NPV().

This part is your independent practice: most formulas are not given. Each step names the in-class cell that used the same method, or gives a pattern to complete. The math column says what each cell must compute.

FILL IN THE YELLOW CELLS

#	Cell	Build it	In math
1	A38	1 Fill to A1037.	$k = 1, \dots, 1000$ trial number
2	D38	<code>=NORM.S.INV(Data_Trials!B2)</code> Fill to D1037.	$z_k = \Phi^{-1}(u_k)$ shared shock
3	E38	<code>=-\$B\$10+NPV(\$B\$7,\$B\$11+\$B\$12*\$D38,\$B\$11+\$B\$12*\$D38,\$B\$11+\$B\$12*\$D38)</code> Fill to E1037.	$NPV_{A,k}$
4	F38	<code>=-____+NPV(____,____,____,____)</code> Fill each _____. Same idea as step 3 above • E38. Fill to F1037.	$NPV_{B,k}$
5	B17	Same method as Part C • B16 . Adapt it to this sheet. Fill to C17.	NPV_A, NPV_B
6	B18	Same method as Part C • B17 . Adapt it to this sheet. Fill to C18.	s_A, s_B
7	B19	Same method as Part C • B18 . Adapt it to this sheet. Fill to C19.	$NPV_{A,5\%}, NPV_{B,5\%}$
8	B20	<code>=COUNTIF(____,____)/COUNT(____)</code> Fill each _____. New here: see "Try it first". Fill to C20.	$P(\text{Loss}_A), P(\text{Loss}_B)$
9	B21	Same method as Part C • B20 . Adapt it to this sheet. Fill to C21.	n_A, n_B must be 1,000
10	B22	<code>=ABS(____)</code> Fill each _____. New here: see "Try it first".	$ NPV_A - NPV_B $ dollars
11	B23	<code>=ABS(____)</code> Fill each _____. Same idea as step 10 above • B22.	$ P(\text{Loss}_A) - P(\text{Loss}_B) $ percentage points
12	B25 B30 B34	B25: 1-2 short paragraphs: recommend an option, cite mean NPV and loss probability with denominators, state one limitation, and name a reversal condition. B30: One line per claim: identify the screening or capital claim, then point to the exact workbook cells or named SPSS output supporting it. B34: Either No AI used, or identify the tool, task, retained output, changes, tests or source checks, and remaining limitations.	

✓ **Done when:** Both options have 1,000 numeric NPVs from fixed Data_Trials inputs, stored-fraction loss probabilities, a written recommendation, evidence map and AI disclosure. **Save.**

⚠ **Most common mistake:** putting the investment *inside* NPV() : Excel would discount it as if it were paid in year 1. Also: calling B23 a "percent change" (a gap between two probabilities is in **percentage points**), and dividing by 1000 instead of COUNT() .

Stuck? Three quick fixes

- **All NPV rows are identical.** D38 must use plain B2 (moves down), and E38/F38 must use \$D38 (column locked, row free).
- **Loss probability is 0% or 100%.** Check the sign: the formula starts with =-\$B\$10+NPV(.
- **Where's the finish check?** Press **Ctrl+G**, type A1040 . The *Check your work* box confirms structure, not your answer.

Optional stretch: does 12% change the answer? (ungraded)

In a *copy* of your finished workbook, change B7 from 8% to 12%. Do the option ranking and the loss-probability gap change? Write two sentences for yourself: one finding and one possible reversal condition. Don't submit the copy.

Submit

Brightspace · LAB02

What to hand in, and how it's graded

Due Tue, Sep 29 · 11:59 PM ET

YOUR EXACT FILENAMES

Upload	Filename	Must contain
Workbook	LAB02_FirstName_LastName_ NetID.xlsx	Every sheet: formulas, interpretations, SPSS captions, evidence map, AI disclosure.
Memo	LAB02_FirstName_LastName_ NetID.pdf	One page, using the template below. A .docx is also accepted; upload one format.
Keep, don't upload	LAB02_FirstName_LastName_ NetID.sps LAB02_FirstName_LastName_ NetID.spv	Your SPSS syntax and output. Upload only if Brightspace asks.

BEFORE YOU SUBMIT: TICK EVERY BOX 0 of 32 done

Files and names

- ☐ Workbook saved as LAB02_FirstName_LastName_
NetID.xlsx
- ☐ Memo exported as LAB02_FirstName_LastName_
NetID.pdf (or .docx ; upload one format only)
- ☐ SPSS files kept as LAB02_FirstName_LastName_
NetID.sps and LAB02_FirstName_LastName_
NetID.spv (upload only if Brightspace asks)
- ☐ NetID is the part before @nyu.edu (like jd1234), not your N-number or full email. Underscores between fields, no spaces.

Part B · IN-CLASS_Lab Part B

- ☐ D15 displays the Modular Hub EMV as **5.40 million dollars**, and D16 agrees with the larger branch EMV. Save the same workbook and resume at IN-CLASS_Lab Part C!A7 .
- ☐ B18 statement chosen
- ☐ B19 written: Reason
- ☐ B23 captions SPSS **H2 Capacity EMV Audit**

Part A · IN-CLASS_Lab Part A

- ☐ B18 displays the conditional churn risk as **35.3%**, using high-ticket complete pairs as the denominator. Save the same personalized workbook and resume at IN-CLASS_Lab Part B!A7 .
- ☐ B20 statement chosen
- ☐ B21 written: Reason
- ☐ B25 captions SPSS **H1 Conditional Churn Risk**

Part C · IN-CLASS_Lab Part C

- ☐ B20 reports **1,000** numeric profit trials, B16:B18 contain mean, sample standard deviation and 5th-percentile profit, and the trial area contains no formula errors. Save the same workbook.
- ☐ B22 statement chosen
- ☐ B23 written: Reason
- ☐ B27 captions SPSS **H3 Reference Profit Summary**

Checkpoint · IN-CLASS_Lab Part C

- ☐ B1033:H1035 contains one evidence-based principle and one reversal condition without claiming that the take-home option has already been selected.
- ☐ B1033 written: Downside-governance checkpoint
- ☐ B1037 captions SPSS **H4 Applied Downside Check**

Whole workbook

- ☐ Every *Check your work* box reads "✓ Checked"
- ☐ No #NAME? , #VALUE! , #REF! , #NUM! or #DIV/0! anywhere
- ☐ Data sheets unchanged; formulas, not pasted values

Upload

- ☐ Both files uploaded to the Brightspace LAB02 assignment by **Tuesday, September 29, 2026, at 11:59 PM ET**
- ☐ Reopened the uploaded files from Brightspace to confirm they open and are the final versions

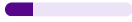
Take-home · POST-CLASS_Part 1

- ☐ Both options have 1,000 numeric NPVs from fixed Data_Trials inputs, stored-fraction loss probabilities, a written recommendation, evidence map and AI disclosure.
- ☐ B25 written: Executive recommendation
- ☐ B30 written: Memo evidence map
- ☐ B34 written: AI disclosure

Memo

- ☐ One page; To, From, Date and Subject filled in; leads with the recommendation
- ☐ Every number has a unit, a denominator and a source cell or SPSS output
- ☐ Separates association from cause, and names what would reverse the recommendation
- ☐ AI disclosure included; every [bracketed prompt] deleted

GRADING MATRIX · 500 POINTS

Dimension	Points	Where it shows up in LAB02
Decision frame, KPI/guardrail, and stakeholder relevance	50 	Memo answers both CFO decisions first; names the guardrail (loss probability, risk tolerance).
Data provenance, grain, quality, and preparation	65 	1,000 rows unchanged; AccountID as text; blanks treated as missing; complete-pair denominator (Part A, SPSS H0).
Method fit and correct required Excel/SPSS implementation	115 	Formulas in Parts A-C and the take-home; SPSS H1-H4 run and captioned.
Assumptions and diagnostic evidence	75 	Probabilities total 100%; 1,000 numeric trials; no formula errors; shared-shock limitation stated.
Uncertainty, effect size, and practical interpretation	70 	35.3% vs. 12.0% baseline; EMV premium; 5th percentile in dollars; gaps in percentage points; association is not cause.
Independent validation, sensitivity, and reasonableness checks	55 	SPSS captions agree with Excel; checks all "✓ Checked"; a reversal condition named.
Recommendation, accessible evidence, and communication	45 	Memo leads with the recommendation; every number has a unit and denominator; evidence map traces each claim.
Reproducibility, sources, privacy, and AI disclosure/validation	25 	Correct filenames; formulas, not pasted values; .sps and .spv kept; AI disclosure present.
Total	500	The rubric applies to the whole submission.

To: Chief Financial Officer and Executive Risk Committee, CloudFlow Analytics
From: [Your name], Senior Business Risk Analyst
Date: [Submission date]
Subject: [Decision-focused subject, e.g. "Recommendation on churn screening and the three-year capital option"]

1. Recommendation 2–4 sentences

[Answer both decisions first. Screening: should CloudFlow prioritize high-ticket accounts, and why? Capital: which option do you recommend, and the main business reason.]

2. Key evidence 4–6 rows

Finding	Value and unit	Denominator	Source
<i>[e.g. churn risk among high-ticket accounts]</i>	<i>[value, %]</i>	<i>[n = ?]</i>	<i>[Part A B18; SPSS H1]</i>
<i>[baseline churn]</i>	<i>[value]</i>	<i>[n]</i>	<i>[cell]</i>
<i>[mean NPV, each option]</i>	<i>[\$]</i>	<i>[trials]</i>	<i>[cells]</i>
<i>[loss probability, each option]</i>	<i>[%]</i>	<i>[trials]</i>	<i>[cells]</i>

3. Data and scope 1–2 sentences

[Which records count (complete pairs; blanks are missing), what population the evidence describes, and why that matters.]

4. Uncertainty and limits 2–3 sentences

[What the evidence can't show: association is not proof screening prevents churn; EMV and mean NPV are averages, not guarantees; the shared-shock assumption. Add one alternative explanation.]

5. What would change my recommendation 1–2 sentences

[A specific reversal condition, e.g. a stricter loss limit or a different discount rate, and the next evidence or check you'd want.]

AI disclosure: *["No AI used." or the tool, task, what you kept, what you changed, and how you checked it.]*

⚠ Delete every [bracketed prompt] before submitting. Cite the key numbers; don't paste the evidence map into the memo.

Optional practice (not graded)

Everything here is optional and ungraded. Each topic asks one question and answers it in Excel, then SPSS, then Python, then R, so you can see how the same idea looks in each tool. Topic 1 reproduces the in-class Parts A-C and stops before the take-home; the other topics go one step beyond the lab. Run any code in the folder that holds `cloudflow_churn_cohort.csv` and `reference_trials.csv`. Start after your workbook and memo are done.

Companion notebooks in your package:

- `LAB02_Python_Exploratory.ipynb`
- `LAB02_R_Exploratory.ipynb`

Topic 1 · Reproduce Parts A-C Reproduce

Same question in Excel → SPSS → Python → R.

Excel Parts A, B, C

Your workbook **is** the Excel version. Reopen `IN-CLASS_Lab Part A`, `B` and `C` and confirm every *Check your work* box reads "✓ Checked". Then change one input in a *copy* (for example the salvage value in Part C) and watch which results move.

SPSS Parts A, B, C

`LAB02_SPSS.sps` already reproduces the parts: H1 is Part A, H2 is Part B and H3 is Part C. Run **Run > All** in a fresh session and compare each output with your workbook. Counts must match exactly and dollar values within \$0.01.

Rebuilds the two-way table and all six probabilities, both EMVs and the 1,000-trial profit summary. Your printed numbers should match your workbook exactly (35.3%, 5.40, and the Part C summaries).

```
# LAB02 Parts A-C in Python (optional, not graded).
# Put this file next to cloudflow_churn_cohort.csv and reference_trials.csv.

import numpy as np
import pandas as pd
from statistics import NormalDist

# ---- Part A: two-way table and conditional churn risk ----
df = pd.read_csv("cloudflow_churn_cohort.csv", dtype=str, keep_default_na=False, na_values=[""])
pairs = df.dropna(subset=["Churn", "HighTickets"]) # complete pairs only; blanks are missing
print(pd.crosstab(pairs["HighTickets"], pairs["Churn"], margins=True, margins_name="Total"))

churn = pairs["Churn"].eq("Yes")
high = pairs["HighTickets"].eq("Yes")
p_c, p_h, p_both = churn.mean(), high.mean(), (churn & high).mean()
print(f"P(C)          = {p_c:.1%}")
print(f"P(H)          = {p_h:.1%}")
print(f"P(C and H)     = {p_both:.1%}")
print(f"P(C or H)      = {p_c + p_h - p_both:.1%}")
print(f"P(H | C)       = {(churn & high).sum() / churn.sum():.1%} <- among churners")
print(f"P(C | H)       = {(churn & high).sum() / high.sum():.1%} <- among high-ticket accounts")

# ---- Part B: payoff table and EMV ($ millions) ----
p = np.array([0.30, 0.50, 0.20]) # high, medium, low demand
payoff = {"Large Hub": np.array([12, 3, -9]), "Modular Hub": np.array([9, 5, 1])}
emv = {name: float(p @ v) for name, v in payoff.items()}
for name, value in emv.items():
    print(f"EMV({name}) = {value:.2f}")
best = max(emv, key=emv.get)
print(f"Rollback: choose {best}; premium = {max(emv.values()) - min(emv.values()):.2f}")

# ---- Part C: 1,000 reference trials ----
u = pd.read_csv("reference_trials.csv")["u_001"].to_numpy()
mu, sigma, Q, price, salvage, cost = 1000, 250, 1000, 100, 15, 40
profit = lambda d: np.minimum(d, Q) * price + np.maximum(Q - d, 0) * salvage - Q * cost
demand = np.array([NormalDist(mu, sigma).inv_cdf(x) for x in u]) # same as Excel NORM.INV(u, mean, sd)
trials = profit(demand)
print(f"Profit at mean demand = ${profit(mu):.0f}")
print(f"Mean simulated profit = ${trials.mean():.0f}")
print(f"Sample SD              = ${trials.std(ddof=1):.0f}")
print(f"5th-percentile profit = ${np.percentile(trials, 5):.0f}") # linear = PERCENTILE.INC
print(f"Flaw-of-averages gap  = ${profit(mu) - trials.mean():.0f}")
print(f"Trials                 = {trials.size:,}")
```

R Parts A, B, C

The same checks in base R. `quantile(type = 7)` matches Excel's `PERCENTILE.INC`, and `qnorm` matches `NORM.INV`.

```
# LAB02 Parts A-C in R (optional, not graded). Base R only.
# Put this file next to cloudflow_churn_cohort.csv and reference_trials.csv.

# ---- Part A: two-way table and conditional churn risk ----
df <- read.csv("cloudflow_churn_cohort.csv", colClasses = "character", na.strings = "")
pairs <- df[!is.na(df$Churn) & !is.na(df$HighTickets), ] # complete pairs only
print(addmargins(table(HighTickets = pairs$HighTickets, Churn = pairs$Churn)))
churn <- pairs$Churn == "Yes"
high <- pairs$HighTickets == "Yes"
pct <- function(x) sprintf("%.1f%%", 100 * x)
cat("P(C)      =", pct(mean(churn)), "\n")
cat("P(H)      =", pct(mean(high)), "\n")
cat("P(C and H) =", pct(mean(churn & high)), "\n")
cat("P(C or H)  =", pct(mean(churn) + mean(high) - mean(churn & high)), "\n")
cat("P(H | C)   =", pct(sum(churn & high) / sum(churn)), " <- among churners\n")
cat("P(C | H)   =", pct(sum(churn & high) / sum(high)), " <- among high-ticket accounts\n")

# ---- Part B: payoff table and EMV ($ millions) ----
p <- c(high = 0.30, medium = 0.50, low = 0.20)
emv <- c(Large = sum(p * c(12, 3, -9)), Modular = sum(p * c(9, 5, 1)))
print(round(emv, 2))
cat("Rollback: choose", names(which.max(emv)), "; premium =", round(max(emv) - min(emv), 2), "\n")

# ---- Part C: 1,000 reference trials ----
u <- read.csv("reference_trials.csv")$u_001
Q <- 1000; price <- 100; salvage <- 15; cost <- 40
profit <- function(d) pmin(d, Q) * price + pmax(Q - d, 0) * salvage - Q * cost
trials <- profit(qnorm(u, mean = 1000, sd = 250)) # same as Excel NORM.INV(u, mean, sd)
dollars <- function(x) paste0("$", format(round(unname(x)), big.mark = ","))
cat("Profit at mean demand =", dollars(profit(1000)), "\n")
cat("Mean simulated profit =", dollars(mean(trials)), "\n")
cat("Sample SD              =", dollars(sd(trials)), "\n")
cat("5th-percentile profit =", dollars(quantile(trials, 0.05, type = 7)), "\n") # type 7 = PERCENTILE.INC
cat("Flaw-of-averages gap  =", dollars(profit(1000) - mean(trials)), "\n")
cat("Trials                =", length(trials), "\n")
```

Topic 2 · How precise is 35.3%? [Extend](#)

Same question in Excel → SPSS → Python → R.

Excel Parts A

Add a 95% interval next to B18 in a *copy* of your workbook. In an empty cell type `=SQRT(B18*(1-B18)/D9)` (the standard error), then `=B18-1.96*<that cell>` and `=B18+1.96*<that cell>`. You should get about 29.6% to 41.0%. Is the whole interval above the 12.0% baseline?

SPSS Parts A

The same formula interval in SPSS syntax, built from the two counts in your H1 table.

New code: paste it at the end of LAB02_SPSS.sps, make sure the setup has run this session, then select only the pasted lines and choose **Run > Selection**.

```
* How precise is 35.3%? A 95% interval from the counts in your H1 table.
* Type the two counts, then compute the rate, its standard error and the interval.
DATA LIST FREE / churned high_ticket.
BEGIN DATA
96 272
END DATA.
* Rate = churned high-ticket accounts / all high-ticket accounts.
COMPUTE p = churned / high_ticket.
* Standard error of a proportion: square root of  $p(1 - p) / n$ .
COMPUTE se = SQRT(p * (1 - p) / high_ticket).
* 95% interval: p plus or minus 1.96 standard errors.
COMPUTE lower = p - 1.96 * se.
COMPUTE upper = p + 1.96 * se.
FORMATS p se lower upper (F6.3).
LIST p se lower upper.
```

Python Parts A

A bootstrap interval instead of a formula: resample the 272 high-ticket accounts 5,000 times and see how much the rate moves. Compare it with the Excel interval.

```
# Extension: how precise is the 35.3%? A bootstrap confidence interval for P(Churn | HighTickets).
import numpy as np
import pandas as pd

df = pd.read_csv("cloudflow_churn_cohort.csv", dtype=str, keep_default_na=False, na_values=[""])
high = df.dropna(subset=["Churn", "HighTickets"]).query("HighTickets == 'Yes'")
churned = high["Churn"].eq("Yes").to_numpy()

rng = np.random.default_rng(2026) # fixed seed: same answer every run
boot = [rng.choice(churned, size=churned.size, replace=True).mean() for _ in range(5000)]
low, high_ci = np.percentile(boot, [2.5, 97.5])
print(f"P(C | H) = {churned.mean():.1%} (n = {churned.size} high-ticket accounts)")
print(f"95% bootstrap interval: {low:.1%} to {high_ci:.1%}")
print("Question: is the interval still well above the 12.0% baseline?")
```

R Parts A

The same bootstrap in base R with `sample()` and `replicate()` .

```
# How precise is 35.3%? Bootstrap the 272 high-ticket accounts 5,000 times.
df <- read.csv("cloudflow_churn_cohort.csv", colClasses = "character", na.strings = "")
high <- df[!is.na(df$Churn) & !is.na(df$HighTickets) & df$HighTickets == "Yes", ]
churned <- high$Churn == "Yes"
set.seed(2026) # fixed seed: same answer every run
boot <- replicate(5000, mean(sample(churned, replace = TRUE)))
ci <- quantile(boot, c(0.025, 0.975))
cat(sprintf("P(C | H) = %.1f%% (n = %d)\n", 100 * mean(churned), length(churned)))
cat(sprintf("95% bootstrap interval: %.1f%% to %.1f%%\n", 100 * ci[1], 100 * ci[2]))
```

Topic 3 · Is the churn pattern more than chance? Extend

Same question in Excel → SPSS → Python → R.

Excel Parts A

In a *copy*, build the table of counts you would **expect** if churn and high tickets were unrelated: each cell is row total × column total ÷ D11 (for B9: =D9*B11/D11). Then =CHISQ.TEST(B9:C10, <expected range>) gives the p-value. How far apart are the observed 96 and its expected count?

SPSS Parts A

Add expected counts, the chi-square test and phi to the H1 crosstab. Compare the Yes/Yes observed and expected counts.

New code: paste it at the end of LAB02_SPSS.sps , make sure the setup has run this session, then select only the pasted lines and choose **Run > Selection**.

```
* Extension: chi-square test of independence for Churn by HighTickets.
* Run H0 first so CohortData is open.
DATASET ACTIVATE CohortData.
CROSSTABS
  /TABLES=HighTickets BY Churn
  /STATISTICS=CHISQ PHI
  /CELLS=COUNT EXPECTED ROW.
```

Python Parts A

The chi-square test computed by hand, so you can see every step: observed, expected, the statistic and the p-value.

```
# Is the churn / high-ticket pattern more than chance? A chi-square test by hand.
import math
import pandas as pd

df = pd.read_csv("cloudflow_churn_cohort.csv", dtype=str, keep_default_na=False, na_values=[""])
pairs = df.dropna(subset=["Churn", "HighTickets"])
observed = pd.crosstab(pairs["HighTickets"], pairs["Churn"])
n = observed.to_numpy().sum()
# Expected count if the two were unrelated: row total x column total / n
expected = pd.DataFrame(
    [[observed.loc[r].sum() * observed[c].sum() / n for c in observed.columns] for r in observed.index],
    index=observed.index, columns=observed.columns)
chi_sq = (((observed - expected) ** 2) / expected).to_numpy().sum()
p_value = math.erfc(math.sqrt(chi_sq / 2)) # exact for a 2 x 2 table (1 degree of freedom)
phi = math.sqrt(chi_sq / n)
print("Observed:\n", observed, "\n\nExpected if unrelated:\n", expected.round(1))
print(f"\nChi-square = {chi_sq:.1f}, p-value = {p_value:.2g}, phi = {phi:.2f}")
print("A small p-value says the pattern is unlikely to be chance. It still does not show that screening prevents churn.")
```

R Parts A

One line in R: `chisq.test()`. A small p-value still says nothing about cause.

```
# Extension: is the churn / high-ticket association more than chance?
df <- read.csv("cloudflow_churn_cohort.csv", colClasses = "character", na.strings = "")
pairs <- df[!is.na(df$Churn) & !is.na(df$HighTickets), ]
tab <- table(HighTickets = pairs$HighTickets, Churn = pairs$Churn)
test <- chisq.test(tab)
print(tab)
print(round(test$expected, 1)) # counts expected if there were no association
print(test)
# A small p-value says the pattern is unlikely to be chance.
# It still does not show that screening would prevent churn.
```

Topic 4 · Which capacity maximizes expected profit? Extend

Same question in Excel → SPSS → Python → R.

Excel Parts C

In a *copy* of your workbook, list capacities 800, 850, ..., 1,400 in an empty column (e.g. H33:H45). In I32 type `=B16`. Select H32:I45, choose **Data > What-If Analysis > Data Table**, leave Row input blank and set Column input cell to `B9`. Which capacity gives the highest mean profit? Compare it with the critical-ratio answer, about 1,135 units.

Compute profit at capacity 1,000 and 1,150 for each of the same 1,000 reference trials. Does the larger capacity raise mean profit? What happens to the spread?

New code: paste it at the end of LAB02_SPSS.sps , make sure the setup has run this session, then select only the pasted lines and choose **Run > Selection**.

** Extension: compare capacity 1,000 with capacity 1,150 on the same 1,000 reference trials.*

```
GET DATA
  /TYPE=TXT
  /FILE='reference_trials.csv'
  /ENCODING='UTF8'
  /ARRANGEMENT=DELIMITED
  /FIRSTCASE=2
  /DELCASE=LINE
  /DELIMITERS=', '
  /QUALIFIER='"'
  /VARIABLES=
    trial F10.0
    u_001 F20.16.
DATASET NAME CapacityTrials.
COMPUTE Demand=IDF.NORMAL(u_001,1000,250).
COMPUTE Profit1000=MIN(Demand,1000)*100+MAX(1000-Demand,0)*15-1000*40.
COMPUTE Profit1150=MIN(Demand,1150)*100+MAX(1150-Demand,0)*15-1150*40.
EXECUTE.
DESCRIPTIVES VARIABLES=Profit1000 Profit1150
  /STATISTICS=MEAN STDDEV MIN MAX.
```

Sweep capacities from 700 to 1,500 on the same 1,000 trials, then compare the best one with the newsvendor critical-ratio formula $Q^* = F^{-1}\left(\frac{p-c}{p-s}\right)$.

```
# Extension: which capacity maximizes expected profit? (the newsvendor problem)
import numpy as np
import pandas as pd
from statistics import NormalDist

u = pd.read_csv("reference_trials.csv")["u_001"].to_numpy()
demand = np.array([NormalDist(1000, 250).inv_cdf(x) for x in u])
price, salvage, cost = 100, 15, 40

def mean_profit(Q):
    return (np.minimum(demand, Q) * price + np.maximum(Q - demand, 0) * salvage - Q * cost).mean()

capacities = np.arange(700, 1501, 25)
results = pd.DataFrame({"capacity": capacities, "mean_profit": [mean_profit(q) for q in capacities]})
best = results.loc[results["mean_profit"].idxmax()]
print(results.to_string(index=False, float_format=lambda x: f"{x:,.0f}"))
print(f"\nBest simulated capacity: {best.capacity:,.0f} units, mean profit ${best.mean_profit:,.0f}")

# Theory check: the critical ratio says order up to the (p - c) / (p - s) quantile of demand.
ratio = (price - cost) / (price - salvage)
print(f"Critical ratio = {ratio:.3f} -> Q* = {NormalDist(1000, 250).inv_cdf(ratio):,.0f} units")
```

The same sweep in base R, with a plot of mean profit against capacity.

```
# Which capacity maximizes expected profit? Same 1,000 trials, capacities 700 to 1,500.
u <- read.csv("reference_trials.csv")$u_001
demand <- qnorm(u, mean = 1000, sd = 250)
price <- 100; salvage <- 15; cost <- 40
mean_profit <- function(Q) mean(pmin(demand, Q) * price + pmax(Q - demand, 0) * salvage - Q * cost)
caps <- seq(700, 1500, by = 25)
profits <- sapply(caps, mean_profit)
plot(caps, profits, type = "b", xlab = "Capacity (units)", ylab = "Mean profit ($)", main = "Expected profit by capacity")
best <- caps[which.max(profits)]
cat("Best simulated capacity:", best, "units, mean profit $", format(round(max(profits)), big.mark = ","), "\n")
ratio <- (price - cost) / (price - salvage)
cat("Critical ratio", round(ratio, 3), "-> Q* =", round(qnorm(ratio, 1000, 250)), "units\n")
```

Topic 5 · Where does the hub decision flip? Extend

Same question in Excel → SPSS → Python → R.

Excel Parts B

In a *copy*, set D7 to `=1-B7-C7` so the probabilities always total 100%, and put `=D14-D15` in an empty cell such as F14. Choose **Data > What-If Analysis > Goal Seek**: set F14 to 0 by changing B7. The answer is the P(high demand) at which both hubs tie.

SPSS Parts B

A small grid of P(high) values with both EMVs side by side. Find the first row where `large_wins` turns to 1.

New code: paste it at the end of `LAB02_SPSS.sps`, make sure the setup has run this session, then select only the pasted lines and choose **Run > Selection**.

```
* Where does the hub decision flip? Keep P(medium) = 0.50 and try P(high) from 0 to 0.50.
DATA LIST FREE / p_high.
BEGIN DATA
0 .05 .10 .15 .20 .25 .30 .35 .40 .45 .50
END DATA.
* P(low) is whatever is left so the three probabilities total 1.
COMPUTE p_low = 0.50 - p_high.
* EMV = sum of probability x payoff ($ millions) for each hub.
COMPUTE emv_large = p_high * 12 + 0.50 * 3 + p_low * (-9).
COMPUTE emv_modular = p_high * 9 + 0.50 * 5 + p_low * 1.
COMPUTE large_wins = (emv_large > emv_modular).
FORMATS p_high p_low (F4.2) emv_large emv_modular (F6.2) large_wins (F1.0).
LIST.
```

Python Parts B

The same grid in Python, plus the exact tipping point from setting the two EMV lines equal.

```
# Where does the hub decision flip? Keep P(medium) = 0.50 and vary P(high demand).
import numpy as np

large, modular = np.array([12, 3, -9]), np.array([9, 5, 1])    # payoffs in $ millions: high, medium, low
p_high = np.round(np.arange(0, 0.51, 0.05), 2)
rows = []
for ph in p_high:
    probs = np.array([ph, 0.50, 0.50 - ph])
    rows.append((ph, probs @ large, probs @ modular))
print("P(high)  EMV Large  EMV Modular")
for ph, el, em in rows:
    print(f"{ph:7.2f} {el:10.2f} {em:12.2f}" + ("    <- Large wins" if el > em else ""))
# Exact: EMV(Large) = 21p - 3 and EMV(Modular) = 8p + 3, so they tie at p = 6/13
print(f"\nTipping point: P(high) = {6/13:.3f}. Today's forecast is 0.30.")
```

R Parts B

Plot both EMVs as $P(\text{high demand})$ changes and find the tipping point exactly with `uniroot`. How far is today's 30% forecast from it?

```
# Extension: when would the Large Hub win? Keep P(medium) = 0.50 and vary P(high demand).
ph <- seq(0, 0.50, by = 0.01)
emv_large <- 12 * ph + 3 * 0.50 + (-9) * (0.50 - ph)
emv_modular <- 9 * ph + 5 * 0.50 + 1 * (0.50 - ph)
plot(ph, emv_large, type = "l", lwd = 2, col = "firebrick",
     ylim = range(c(emv_large, emv_modular)),
     xlab = "P(high demand)", ylab = "EMV ($ millions)", main = "Where the decision flips")
lines(ph, emv_modular, lwd = 2, col = "purple")
abline(v = 0.30, lty = 2) # today's forecast
legend("topleft", c("Large Hub", "Modular Hub"), col = c("firebrick", "purple"), lwd = 2)
tip <- uniroot(function(x) (21 * x - 3) - (8 * x + 3), c(0, 0.5))$root
cat("The Large Hub overtakes the Modular Hub when P(high) exceeds", round(tip, 3), "\n")
```